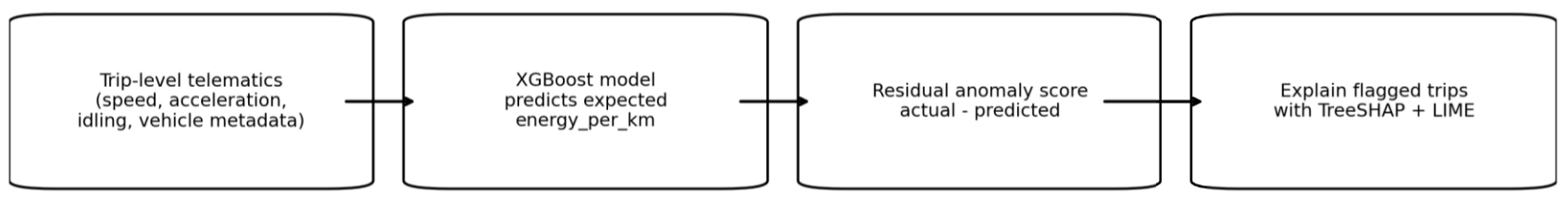
# Explaining Vehicle Energy Anomalies in Telematics

with XGBoost, Manual SHAP, and LIME

# Real-world problem

- Vehicle telemetry is rich, but anomaly scores alone are not actionable
- A flagged trip may reflect route conditions, driving behavior, missing context, or real degradation
- We therefore need both detection and explanation




Energy anomaly detection and explanation pipeline

# Dataset and task

**Dataset VED**

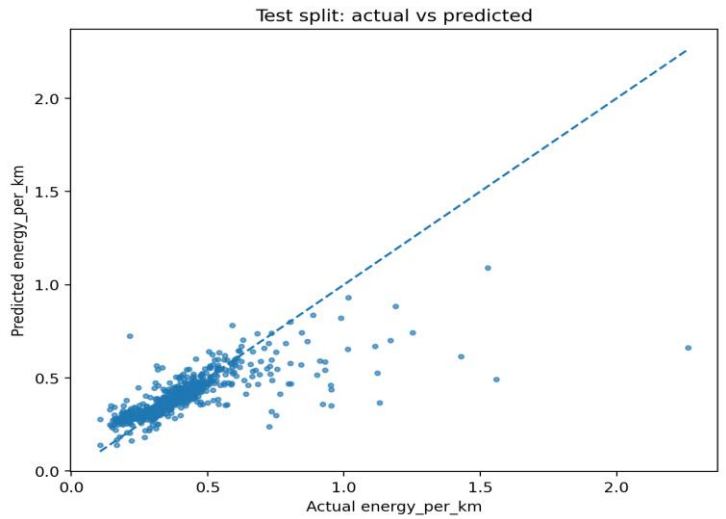
| | |
|---|---|
| **Type of data** | one row – per trip |
| **Target** | energy_per_km |
| **Anomaly score** | positive residual |

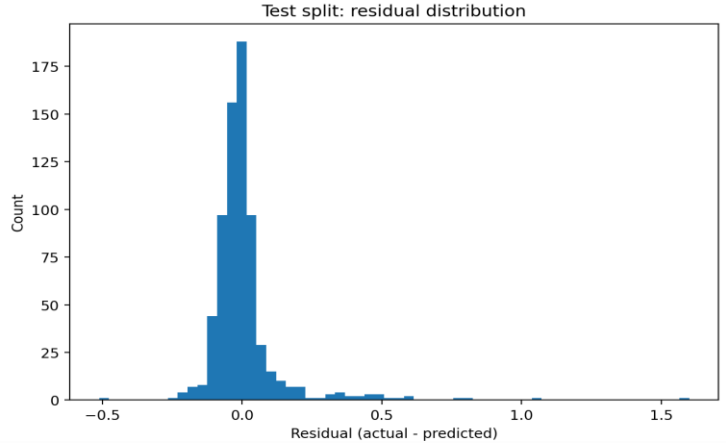
**Main result**

**MAE** = 0.0670, **RMSE** = 0.1332, $\mathbf{R^2}$ = 0.5377 on the test split

# Black-box model

- XGBoost regressor
- strong baseline
  for tabular telematics features
- residuals define anomalies



# XAI methods

## Manual SHAP

Shapley-style attribution implemented from first principles

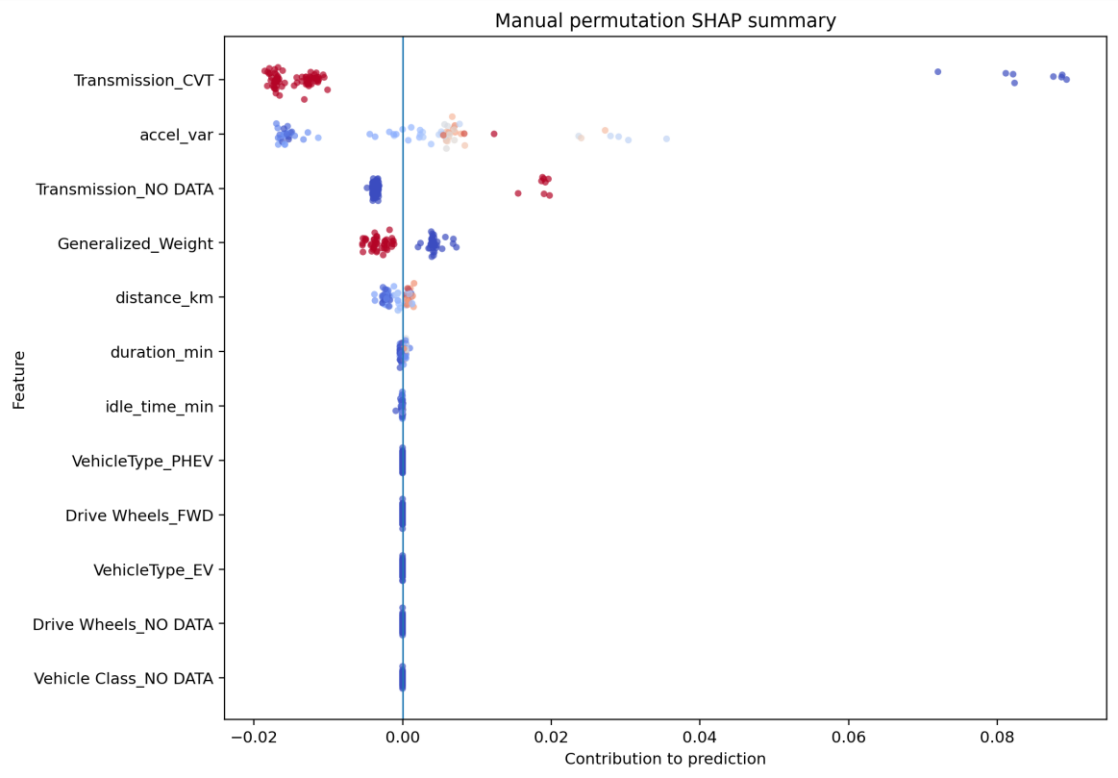
## LIME

local surrogate baseline

## Goal

compare global and local interpretability

# Global explanation

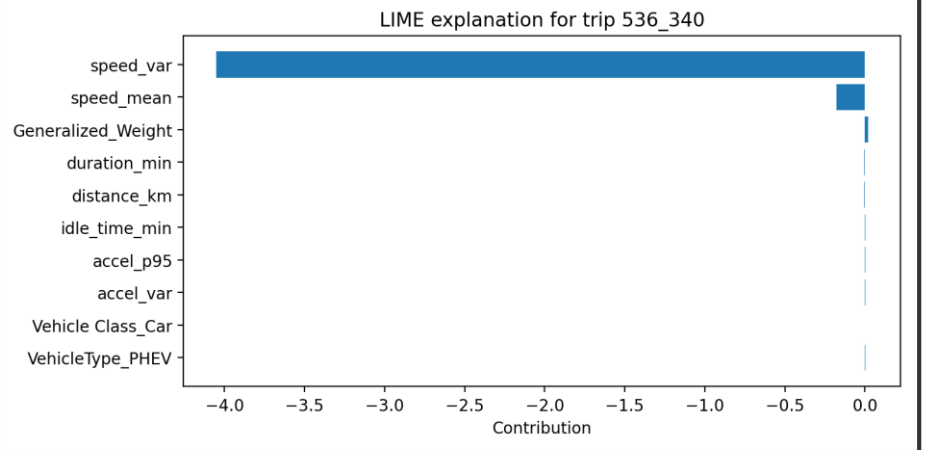


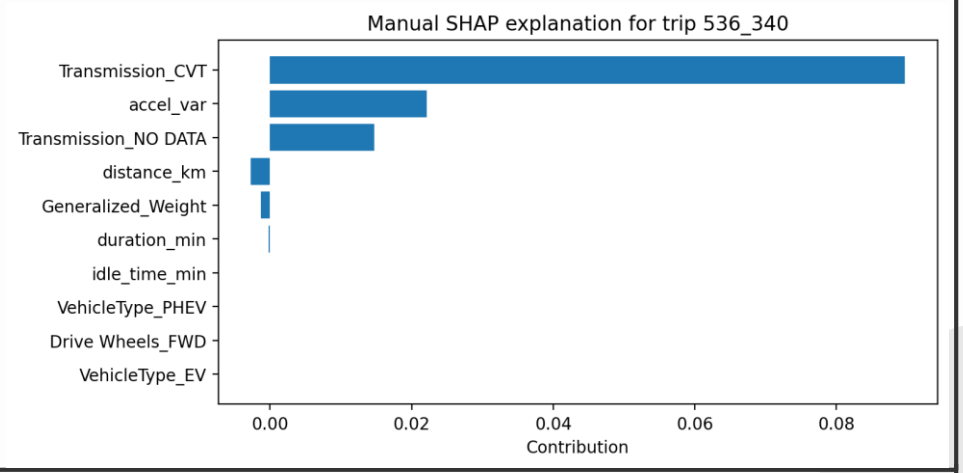

- transmission indicators
- acceleration variance
- generalized weight
- distance and duration

# Local case study: good explanation

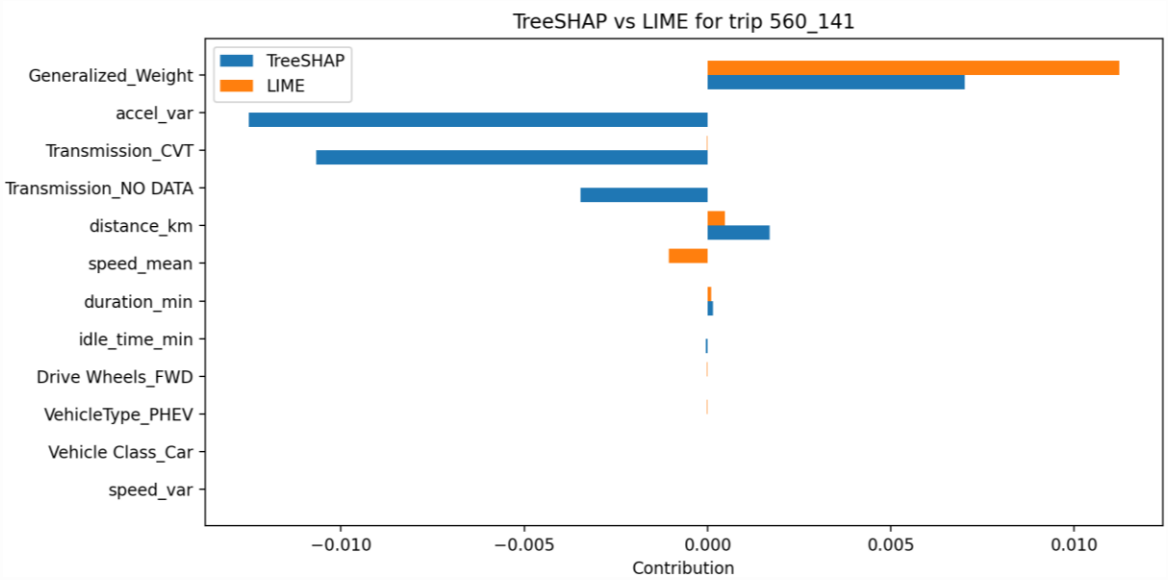
Use trip **536_340**:

- strong anomaly
- local LIME fit is strong
- compare main SHAP and LIME signals



# Global explanation



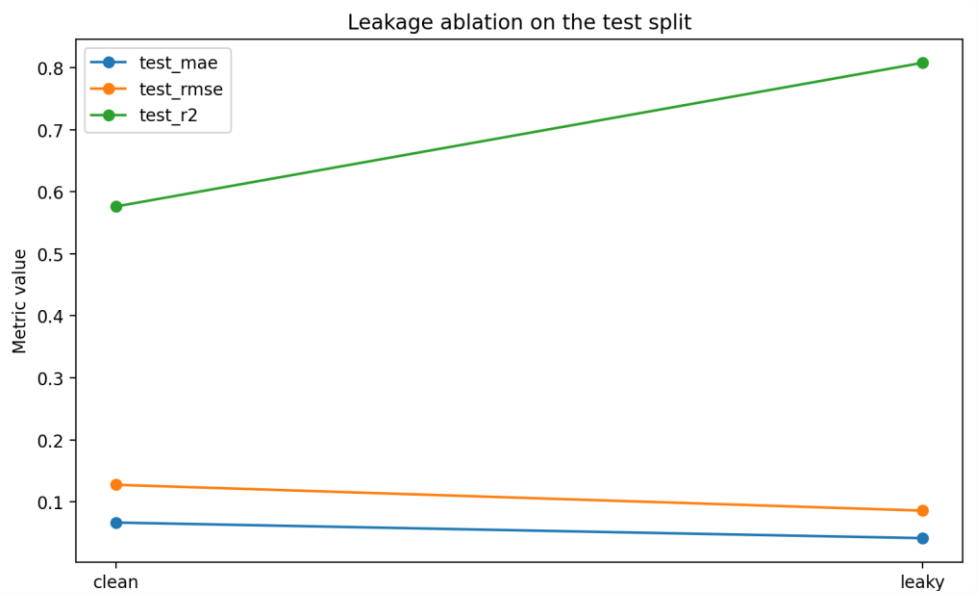

Use trip **560_141**:

- largest residual (test split)
- but LIME local $R^2$ is only 0.101
- explanation readability does not guarantee faithfulness

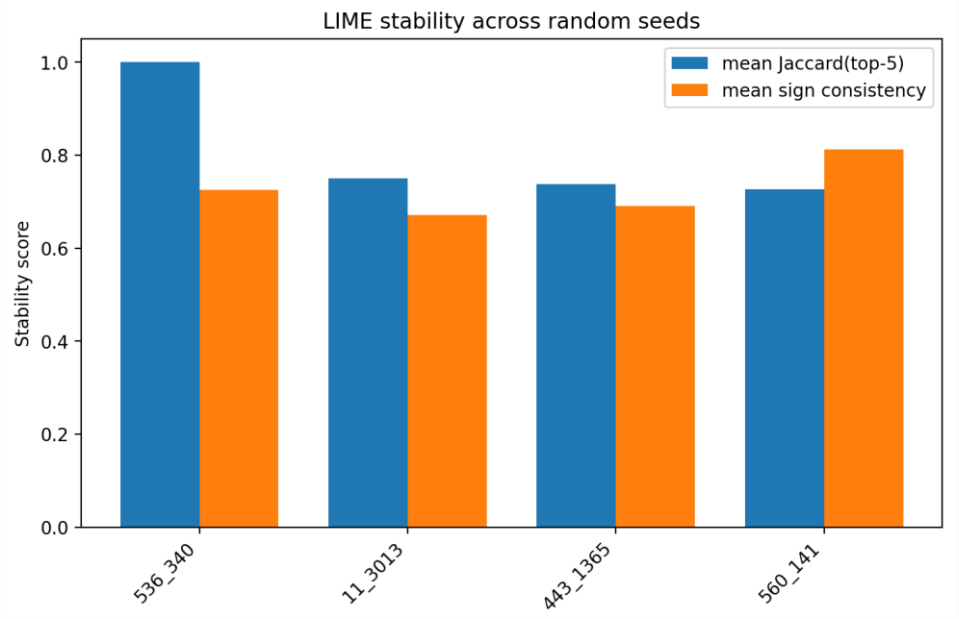
# Trustworthiness audit

Three strongest findings

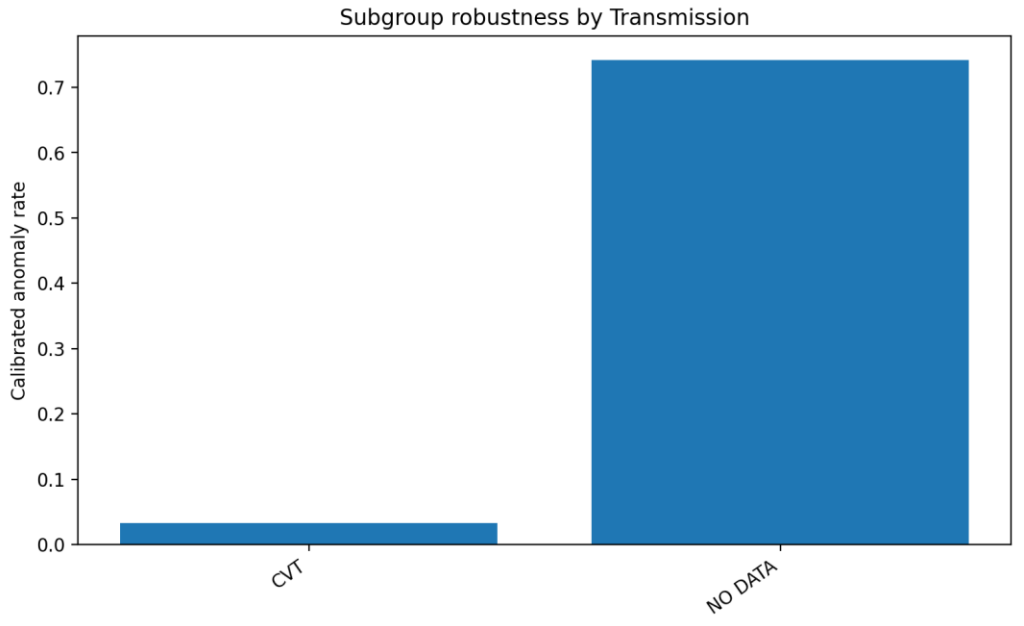
# 01 Leaky model looks much better than the cleaned model



# 02 LIME stability is case-dependent



# 03 Subgroup performance differs sharply for Transmission = NO DATA



# Conclusions

- Anomaly scores need explanations
- SHAP and LIME provide useful but different views
- High apparent performance can be misleading under leakage
- Trustworthy XAI requires auditing, not just plotting

# Thank you for attention!